

1. Define query language.

Query language is computer language which used to make queries in databases and information system.

It is also known as data query language.

Eg: structured query language.

2. What are the types of Query language?

DDL or Data Definition Language

DQL or Data Query Language

DML or Data Manipulation Language

DCL or Data Control Language

TCL or Transaction Control Language

3. Define UML.

UML - Unified Modeling language.

It defines a standard set of modeling diagrams for all stages of developing a

software system.

This information describes the entity-relationship model of database design.

The Object Management Group is a consortium that created the UML standard.

4. Define Hashing

Hashing is a DBMS technique for searching for needed data on the disc without utilising an index structure.

The hashing method is basically used to index items and retrieve them in a database since searching for a specific item using a shorter hashed key rather than the original value is faster.

Two types of hashing:

Static hashing

Dynamic hashing



5. Define Static and Dynamic hashing

Static hashing

It is a hashing technique that enables users to lookup a definite data set. Meaning, the data in the directory is not changing, it is "Static" or "fixed".

In this hashing technique, the resulting number of data buckets in memory remains constant.

Dynamic Hashing

It is a hashing technique that enables users to lookup a dynamic data set. Means, the data set is modified by adding data to or removing the data from, on demand hence the name 'Dynamic' hashing. Thus, the resulting data bucket keeps increasing or decreasing depending on the number of records.

In this hashing technique, the resulting number of data buckets in memory is ever-changing.



b. Define Strong and Weak Entity sets.

Strong Entity sets

A strong entity is not dependent on any other entity in the schema. A strong entity will always have a primary key.

Strong entities are represented by a single rectangle.

The relationship of two strong entities is represented by a single diamond.

Various strong entities, when combined together, create a strong entity set.

Weak Entity sets:

A weak entity is dependent on a strong entity to ensure its existence.

Unlike a strong entity, a weak entity does not have any primary key.

It instead has a partial discriminator key. A weak entity is represented by a double rectangle.

The relation between one strong and one weak entity is represented by a double diamond.

This relationship is also known as identifying relationship.



7. Define View in SQL

View is a virtual table and the changes can be reflected to the original relation.

It is derived from original name.

A view contains rows and columns, just like a real table.

Application of DBMS:

Railway Reservation system

Library Management system

Banking

Education sector

Credit card exchanges

Social Media sites

Broadcast communications

8. Define primary key.

A primary key is the column or columns that contain values that uniquely identify each row in a table.

A database table must have a primary key for option to insert, update, restore, or



delete data from a database table.

9) Define Foreign Key

A foreign key is a column or columns of data in our table that refers to the unique data values often the primary key data -- in another table

Foreign keys link together two or more tables in a relational database.

10) Define relational database system.

A relational database is a collection of information that organizes data in predefined relationships where data is stored in one or more tables (or "relations") of columns and rows, making it easy to see and understand how different data structures relate to each other. Relationships are a logical connection between different tables, established on the basis of interaction among these tables.



2 type of DML

High level or Non procedural DML

Low level or Procedural DML

11) Define referential integrity.

This integrity is between 2 table if a foreign key in table 1 refers to the primary key of table 2 then every value of the foreign key in table 1 must not be null and be available in table 2.

12. Define B tree and B⁺ tree

B tree

B tree is a popular terminology that belongs to the computer science family. It is a balancing tree that helps in maintaining and sorting data, and also grants searches, insertions, deletions and sequential access. It is also known as an m-way tree where m



is used to determine the order of the tree. Depending upon the value of m , a B-tree node can have more than one key and more than two children based on the value of m .

B⁺ tree

B⁺ tree is nothing but an advancement of B tree that allows efficient and smooth insertion, deletion, and sequential access. It is also known as ~~n-array tree~~ which holds a large number of children per node.

B⁺ tree helps in reducing the drawback faced by B tree by saving the data pointers at the leaf node level and simply storing the key values in the internal nodes.



13) Define Boyce Codd Normal form

BCNF (Boyce Codd normal form) is an advanced version of the third normal form (3NF), and often, it is also known as the 3.5 Normal form. A relation is said to be in BCNF in DBMS if the relation is already in 3NF, and also, for every functional dependency (say, $x \rightarrow y$), x is either the super key or the candidate key.

14) Define Buffer management

A buffer manager in DBMS is in charge of allocating buffer space in the main memory, so that the temporary data can be stored there.

The buffer manager sends the block address if a user requests certain data and the data block is present in the database buffer in the main memory.



15) Define Multiple granularity

It can be defined as hierarchically breaking up the database into blocks which can be locked.

The multiple granularity protocol enhances concurrency and reduces lock overhead.

It maintains the track of what to lock and how to lock.

It makes easy to decide either to lock a data item or to ~~unlock~~ a data item. This type of hierarchy can be graphically represented as a tree.

16) Define ACID

In transaction processing, ACID (atomicity, consistency, isolation and durability) is an acronym and mnemonic device used to refer to the four essential properties a transaction should possess to ensure the integrity and reliability of the data involved in the transaction.



17) Define Database

A database is an organized collection of structured information or data typically stored electronically in a computer system.

A database is usually controlled by a database management system (DBMS).

The data can then be easily accessed, managed, modified, updated, controlled and organized.

Most databases use structured query language (SQL) for writing and querying data.

18) Define Functional Dependencies

The functional dependency is a relationship that exists between two attributes. It typically exists between the primary key and non key attribute within a table.

$$X \rightarrow Y$$



The left side of functional dependency is known as a determinant, the right side of the production is known as a dependent.

19) Normalization

Normalization is the process of organizing data in a database.

It includes creating tables and establishing relationships between these tables according to rules designed both to protect the data and to make the database more flexible by eliminating redundancy and inconsistent dependency.

20) Define Data model

A data model in Database management system (DBMS) is the concept of tools that are developed to summarize the description of the database.



Data models provide us with a transport picture of data which helps us in creating an actual database.

It shows us from the design of the data to its proper implementation of data.

21) Define Fuzzy checkpoint

An alternative to traditional cache consistent checkpoints involving less overhead but may require more restart work.

In a fuzzy checkpoint, the recovery manager only writes the buffer pages since the previous checkpoint. Most of these pages should have already been written to disk before the checkpoint.

At restart time, the recovery manager uses the most recent fuzzy checkpoint records in the log.



22) Write the SQL commands for creating the table.

SQL create Table Syntax: CREATE TABLE table_name (column datatype, column2 datatype, ... column-N datatype, PRIMARY KEY (one or more column)); CREATE TABLE is the keyword to tell the database to create a table.

23) Define Atomicity

Atomicity in DBMS refers to the property of a database transaction where all the actions within the transaction are executed as a single, individual unit of work.

It ensures that the transaction is either fully completed or fully rolled back to the state it was in before the transaction began.



24. Define Serializability.

Serializability is a concept in database management systems (DBMS) that ensures the correctness of concurrent transactions.

Serializability is a way to check if the execution of two or more transactions are maintaining the database consistency or not.

25. Define Lock and its uses.

Locking is the process by which the DBMS restricts access to a row in a multi-user environment.

It helps in eliminating the concurrency problem by locking a particular transaction to a particular user.

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